



Exploring Historical Routes and Waypoints with MICOLL Digital Map

Fabio Giachelle^{1,2} , Jake Dyble² , Giorgio Maria Di Nunzio¹ ,
and Stefania Gialdroni²

¹ Department of Information Engineering, University of Padua, Padua, Italy
{fabio.giachelle,giorgiomaria.dinunzio}@unipd.it

² Department of Private Law and Critique of Law, University of Padua, Padua, Italy
{jacobarthur.dyble,stefania.gialdroni}@unipd.it

Abstract. This work presents the MICOLL digital map, a web-based application for searching for historical routes and waypoints in the context of the ERC project “Migrating Commercial Law and Language (MICOLL). Rethinking Lex Mercatoria”. The MICOLL digital map is an ongoing project that aims to allow historians and experts to easily search and discover the historical routes, used for exchanging goods and information, that connected European cities during the 11th-17th centuries. The final goal is to show their evolution over time by linking historical information to spatial waypoints and enabling a visual and interactive consultation over a digital map.

Keywords: Historical Routes · Digital Maps · Cultural Heritage · Linking Historical Evidence

1 Background

Migrating Commercial Law and Language (MICOLL)¹ is an ERC-funded interdisciplinary project involving experts from different domains (historians, jurists, linguists, and computer engineers). The project aims to evaluate and reconstruct the migration of terms, specific for the legal domain, across Europe [4] with a focus on the geographic areas comprehending Germany and Northern Italy. Specifically, the project aims to examine the routes used by merchants to transfer goods and written information, in order to investigate the existence of a shared commercial law (i.e., “lex mercatoria”) and thus determine whether a common language was spoken by medieval merchants with respect to commercial law. Despite the fact that the “lex mercatoria” is likely to be a myth to support contemporary arguments for private-ordering solutions [3, 7], there is historical evidence supporting the migrations of terms from cities of the Italian peninsula (e.g., Florence, Genoa, and Venice) to Germany and Northern Europe thanks

¹ <https://www.micoll-erc.eu>.

to the Trans-Alpine tract, which is regarded as a key waypoint for European itineraries [10].

The MICOLL map is one of the key outputs of the project, enabling experts to search for historical routes consisting of multiple waypoints and consult the related proofs and evidence supporting each route/waypoint type by providing historical attestations (i.e., the century, year or date of attestation). In this regard, the map is intended not only to support social historians comprehending whether and how commercial terms themselves might have migrated but also for studying mobility and communication phenomena.

Despite the importance of having reliable digital historical maps, existing software and tools still are far from the level of accuracy required to be used confidently by historians and experts for research purposes. Yet, they are intended mainly for dissemination and public engagement activities. Indeed, several projects aim at digitalizing a limited number of sources, including for instance the *DECIMA* project [15], which focuses on creating a digital map tailored for the city of Florence during the renaissance period. In contrast, the prominent *ORBIS* [13] project exploits several different sources with the aim of achieving a good approximation of the historical travel routes in the Roman Empire trying to estimate both time and cost of moving between two points of the Empire.² Another major project is *EmDigIt* which aims to create a navigation system for the Early Modern Period.³

To properly design and implement digital historical maps several aspects need to be taken into account: (i) support multiple map layers and overlays such as digital scans of historical maps as layers over a base tiled layer [9, 12]; (ii) store data into a historical geographic database [8]; (iii) use Linked Open Data (LOD) approaches and embrace the Findable, Accessible, Interoperable, Reusable (FAIR) principles [5, 11] and (iv) storytelling, user engagement and increase the dialogue connecting past, present, and future [1, 6].

Another important aspect is the capability of visually explaining how a map changed over time by linking points of interest and their properties to the supporting historical evidence and sources. However, current digital historical maps do not provide this kind of feature. In this regard, the MICOLL map will try to fill this gap by providing features such as (i) support for routes and waypoints on both land and water; (ii) show their change over time; (iii) integrate multiple different evidence sources supporting informative aspects of the map and enabling users to consult and cite them over structured and spatial layers. Despite the MICOLL Map is currently under construction, some key aspects are already implemented and available to the public.

2 MICOLL Map

The MICOLL map is publicly available as a web-based application at <https://w3id.org/micoll-map>. It can be consulted without limitations by unregistered

² <https://orbis.stanford.edu>.

³ <https://tinyurl.com/early-modern-digital-itineraries>.

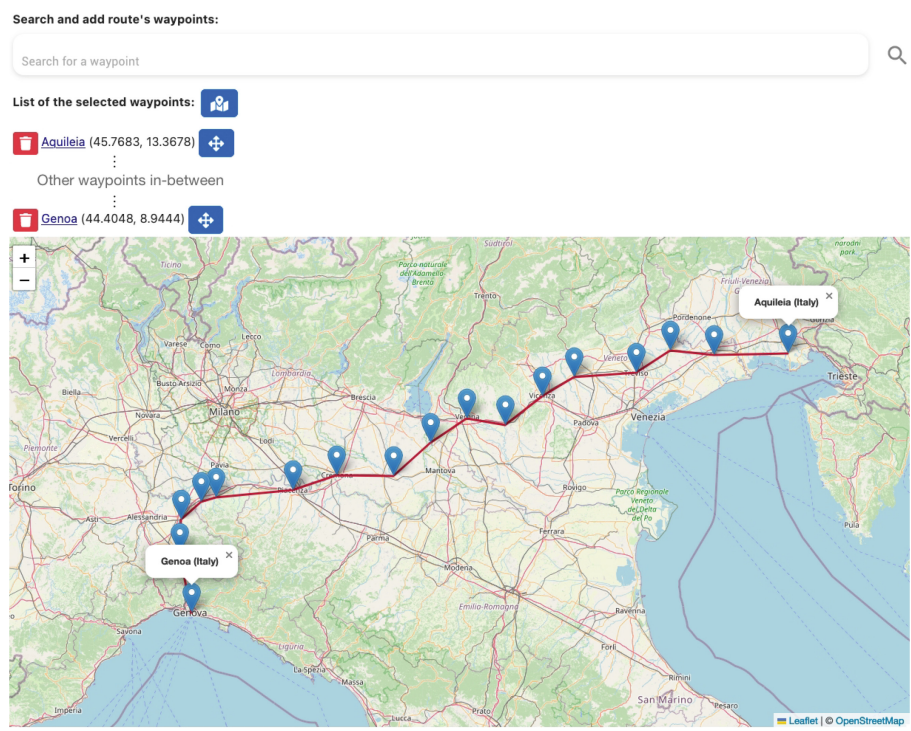


Fig. 1. Interface for editing a route such as the Postumia Road. It is an ancient Roman road connecting the two cities of Genoa and Aquileia. It was a key element of connection between two of the major ports of the Italian peninsula.

Manage Routes				
Name	Description	Route types	Route sources	Actions
Filter...	Filter...	Filter...	Filter...	
Adige River	Nalies to the Adriatic Sea (including Venice)	Waterway	Conati-Arti e mestieri sull'adige: dalle	View Edit Delete
Augsburg-Antwerp	The postal route from Augsburg to Antwerp	Postal route	Krüger-Das älteste deutsche routenth	View Edit Delete
Augsburg-Frankfurt am Main	Augsburg to Frankfurt am Main via Heidelbe	Postal route	Krüger-Das älteste deutsche routenth	View Edit Delete
Augsburg-Genoa	Augsburg to Genoa via Milan	Postal route	Mayr-Wegbüchlin (1612) der fürneme	View Edit Delete
Augsburg-Linz	Augsburg to Linz via Landshut	Postal route	Krüger-Das älteste deutsche routenth	View Edit Delete
Augsburg-Nuremberg	Augsburg to Nuremberg	Postal route	Mayr-Wegbüchlin (1612) der fürneme	View Edit Delete

Fig. 2. List of the routes, recorded in the database, presented in tabular form. The records can be filtered according to the values specified in the column filters. Users can visualize all the information concerning a route of interest by clicking on the *View* button. The name of each route is a link to the corresponding landing page. The registered users can also edit the information concerning each route, or delete it.

users, though a registered account is required (activated upon request) in order to contribute and edit the contents of the platform. Users can access several information including all the current routes and waypoints stored in the database. Specifically, users can access not only the information associated with routes and waypoints but also the associated sources, together with the historical attestations that support each route/waypoint type and their authors. The MICOLL map provides several features to manage different resources, that is, waypoints, routes, sources and their authors. A route is intended as a path connecting multiple waypoints (at least two) supported by some authoritative sources providing information such as route/waypoint types. Users can create a new route by clicking on the *Create a new route* menu item. The interface allows the users to specify multiple information about the route such as its name, a description, its types (e.g., *Postal route*, *Trade route*), evidence sources, and the waypoints. To support the users in the process, the interface allows them to search for waypoints and filter them according to multiple fields. Then users can sort and rearrange the waypoints to create a route, by taking advantage of drag-and-drop facilities. The map automatically updates the route as users add new waypoints, resulting in increased precision and refinement with a higher number of specified waypoints. The default waypoints available are provided by the GeoNames⁴ database and, specifically, by the open dataset containing worldwide cities with a population greater than five hundred people, which is publicly available at <https://public.opendatasoft.com/explore/dataset/geonames-all-cities-with-a-population-500>. In addition, users can define new waypoints by specifying geospatial information such as latitude and longitude. This feature is particularly useful when referring to historical places whose location could have shifted over time. Figure 1 shows the interface for searching, adding, and sorting the waypoints making up a route, in the example is reported the *Postumia Road*, an ancient route connecting Genoa to Aquileia. Users can search for the waypoints of interest using multiple languages since for each waypoint the alternative names, in the different languages, are stored in the database. The alternative names for the waypoints are provided by the GeoNames database.

The map allows the users to inspect each waypoint by clicking on it, so that they can check the information describing the waypoint, the associated types and the supporting sources. Once a route is created, it is listed, in tabular form, among the stored routes so that registered users can manage and further edit it. Figure 2 shows the table reporting the routes stored, so that users can search, filter, and edit them using the action buttons available. In addition, all the information available about a route is summarized in a dedicated landing page, so that users can easily share it using the usual button, as shown in Fig. 3.

Similarly, a specific landing page is also provided for each waypoint, as shown in Fig. 4, and this is integrated with several web mapping services (i.e., OpenStreetMap, Google Maps, Bing Maps, GeoHack, Geonames) through links to the corresponding entry of the waypoint for each of the services considered. Although the Geonames database is quite comprehensive, it does not contain all the way-

⁴ <https://www.geonames.org>.

Adige River

Route name
Adige River

Share route

Nalles to the Adriatic Sea (including Venice) along the River Adige

The movement of goods along the Adige in the medieval period is attested in written documents as early as 845 (in that case from Trento to Verona) (Conati, 2021, p.32). The river could be navigated with rafts from the villages of Terlano and Nalles downstream until Egna Neumarkt where they were transported on larger vessels. Egna Neumarkt was the highest point that could be reached travelling

Route types:

Specify a type for the Route (if applicable)

Waterway

Specify the attestation type (e.g., date, year, century)

Century

Specify the attestation (Century)

XXI

Specify the supporting evidence/source (if available)

Conati-Arti e mestieri sull'adige: dalle valli tirolesi all'adriatico

Specify the supporting evidence/source (if available)

Turri-Lo spazio atesino

Fig. 3. Landing page (excerpt) for the route *Adige River* which describes a route for the movement of goods along the river Adige. The interface presents the route name, a description, and the associated route types (i.e., *Waterway*) along with the historical attestation (i.e., century XXI) and the supporting source (i.e., “Lo spazio atesino” authored by E. Turri).

points of interest; some, like *Cavanella d'Adige*, have been manually added to the MICOLL map. The URL associated with the landing page of a resource (e.g., route, waypoint) is a permanent link (e.g., <https://w3id.org/micoll-map/route/Augsburg-Antwerp>) to ensure that it remains consistent and accessible over time, thus fostering citation and references. Technically, we implemented permanent URLs by relying on the permanent URL redirection service run by the W3C Permanent Identifier Community Group⁵. The web application has been implemented using React.js for the front-end and Django, a Python-based web framework, for the back-end. The base map layer is implemented using the

⁵ <https://www.w3.org/community/perma-id/>.

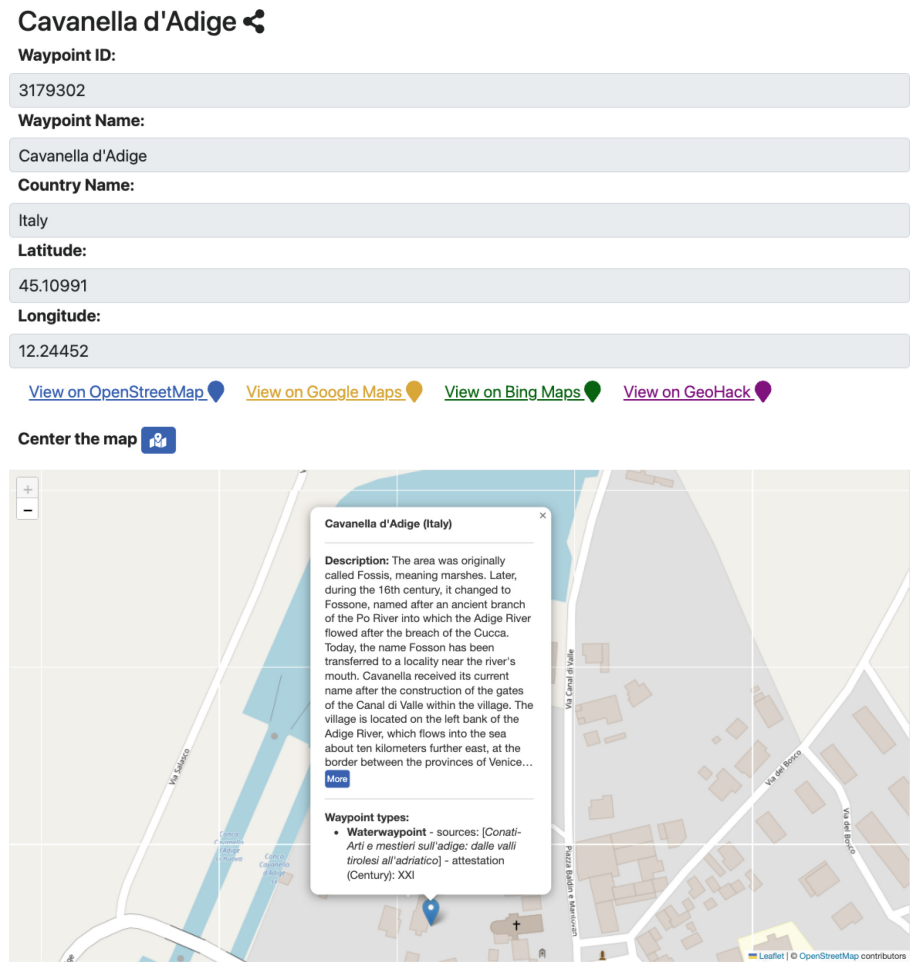


Fig. 4. Landing page (excerpt) for the waypoint *Cavanella d'Adige* which is part of a waterway along the river Adige. The interface presents the waypoint name, the geospatial information (e.g., latitude, longitude), a description, and the associated waypoint types (i.e., *Waterwaypoint*) along with the historical attestation (i.e., century XXI) and the supporting source (i.e., “Arti e mestieri sull’adige: dalle valli tirolesi all ‘adriatico” authored by G. Conati).

Leaflet mapping library and it relies on the OpenStreetMap (OSM) project. For what concerns the database chosen for storing geographic data is PostgreSQL integrated with the PostGIS extension. Figure 5 shows the Entity-Relationship Diagram (ERD) which depicts the entities and the relationships between them that are implemented within the database. Due to space constraints, only the most significant portion of the diagram is illustrated (e.g., only the key entities and attributes are represented). For instance, we can consider the *Waypoint*

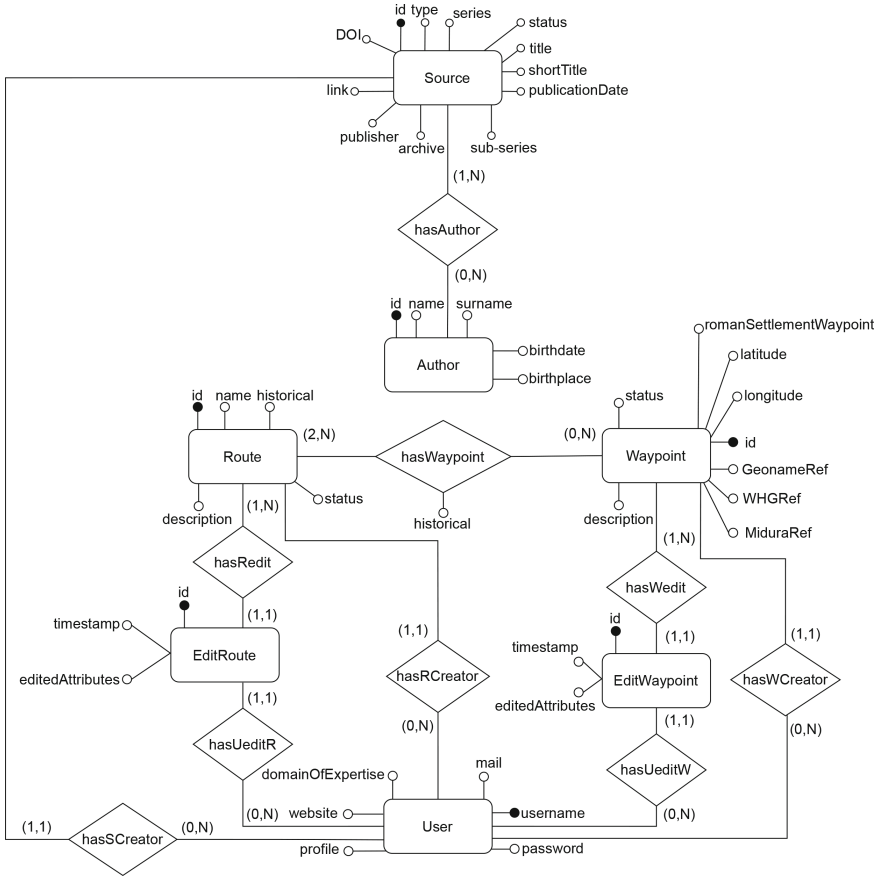


Fig. 5. ERD (excerpt) modelling the database associated with the MICOLL Digital Map. The entities in the diagram are represented as rectangles with rounded angles (e.g., *Waypoint*, *Route*, *Source*) whereas the relationships between the entities are represented as rhombuses (e.g., *hasWaypoint*, *hasAuthor*). The attributes composing the primary identifiers are represented with filled circles (e.g., *username*). The complete ERD is available in the GitHub repository under the `docs` directory.

entity that has several attributes (e.g., latitude and longitude) to represent the associated geospatial information. According to the ERD, a route is always made up by at least two waypoints as indicated by the minimum cardinality (2) in the relationship between the entities *Route* and *Waypoint*. Moreover, it is evident that *Route* and *Waypoint* may have multiple types, as indicated by the capital *N* for the maximum cardinality. One of the most important aspects is the possibility of having evidence sources supporting historical facts (e.g., the Postumia Road was a major trade route in northern Italy). In this regard, the entity *Source* represents the evidence sources, both primary and secondary, providing a historical attestation for a given fact of interest. This is achieved by connecting the

entity *Source* with the entities *RouteCouple* and *WaypointCouple*. *RouteCouple* represent an association between a *Route* and a specific *RouteType*. Similarly, *WaypointCouple* represent an association between a *Waypoint* and a specific *WaypointType*. For what concerns the users of the MICOLL map platform, each user has a *profile* and a *domainOfExpertise* as describing attributes that qualify the expert, for instance, when occurs a modification of the information related to the entities (e.g., *Route*, *Source*, *Waypoint*). In this regard, each update is also recorded within the entities *EditRoute*, *EditWaypoint*, and *EditSource*, where the edited attributes for a resource are tracked along with the temporal information (in the form of a timestamp), hence it is possible to check which user did the last update of the information concerning each entity and when.

The source code of the web application is publicly available on GitHub⁶ and since it is structured to be flexible, users can reuse it in similar contexts where it is necessary to organize and handle data concerning historical itineraries as well as providing search and filtering capabilities.

3 Conclusions

We presented the MICOLL map, a web-based platform designed to allow historians to search, access, and inspect historical routes and waypoints in terms of the supporting historical evidence. Since the MICOLL map is under construction, we plan to enhance it with multiple additional features. Firstly, we plan to integrate the OpenLayers tiling system to better support different stacked layers at once (this is useful for instance to overlay a historical digitized map over a base layer describing the map at present). Secondly, we plan to implement automatic procedures for data ingestion and sharing, in order to enhance interoperability with other existing tools used by historians and experts. In this regard, we plan to provide support for the major geospatial formats (e.g., GeoJSON, Well-Known Text (WKT), and Keyhole Markup Language (KML)). Finally, we plan to improve the searching facilities providing support for searching routes, including both land routes and waterways, connecting two given waypoints in a well-defined timeframe specified by the user.

In terms of collaborations with other national initiatives and projects that have studied and proposed solution to visualize the past of the Italian territory, we would like to analyze the possibility to integrate the MICOLL maps with 3D reconstructions, like The Digitales Forum Romanum [14], or the data and indicators useful for understanding the identities and interactions between local communities, cultural heritage, and landscapes, like the research conducted by [2].

Acknowledgements. This article was written with funding from the European Research Council (ERC) under the European Union’s Horizon 2020 research and innovation programme (Grant agreement no. 101002084).

⁶ <https://github.com/MICOLL-ERC/micoll-erc-map-pub>.

References

1. Basaraba, N.: The emergence of creative and digital place-making: a scoping review across disciplines. *New Media Society* **25**(6), 1470–1497 (2023). <https://doi.org/10.1177/14614448211044942> publisher: SAGE Publications
2. Colaprice, V., Barbuti, N.: Una nuova mappatura digitale per i borghi delle aree interne. *Umanistica Digitale* **15**, 25–46 (Jul2023). <https://doi.org/10.6092/issn.2532-8816/15612>, <https://umanisticadigitale.unibo.it/article/view/15612>, number: 15
3. Cordes, A.: The search for a medieval lex mercatoria. *Oxford U Comparative L Forum* **3** (2003). <https://ouclf.law.ox.ac.uk/the-search-for-a-medieval-lex-mercatoria/>
4. Di Nunzio, G., et al.: “Stick to the Word you Can Afford”. An ERC Project: Migrating Commercial Law and Language (MICOLL) (short paper). In: Damiano, R., Ferilli, S., Striani, M., Silvello, G. (eds.) *Proceedings of the 1st Workshop on Artificial Intelligence for Cultural Heritage*. CEUR Workshop Proceedings, vol. 3286, pp. 44–49. CEUR, Udine, Italy (Nov 2022). https://ceur-ws.org/Vol-3286/#05_paper, ISSN: 1613-0073
5. Gkadolou, E., Prastacos, P.: Historical Cartographic Information for Cultural Heritage Applications in a Semantic Framework. *Cartographica: Int. J. Geograph. Inform. Geovisualization* **56**(4), 255–266 (2021). <https://doi.org/10.3138/cart-2021-0002>, <https://www.utpjournals.press/doi/full/10.3138/cart-2021-0002>, publisher: University of Toronto Press
6. Guntarik, O., Davies, H., Innocent, T.: Indigenous cartographies: pervasive games and place-based storytelling. *Space and Culture* (Mar 2023). <https://doi.org/10.1177/12063312231155348>, publisher: SAGE Publications Inc
7. Kadens, E.: The myth of the customary law merchant. *Texas La Rev.* **90**, 1153–1206 (2012). <http://texaslawreview.org/wp-content/uploads/2015/08/Kadens-90-TLR-1153.pdf>
8. Li, S.: A Multi-level grid database for protecting and sharing historical geographic urban data: a case study of shanghai. *ISPRS Int. J. Geo Inf.* **12**(8), 325 (Aug2023). <https://doi.org/10.3390/ijgi12080325>, <https://www.mdpi.com/2220-9964/12/8/325>, number: 8 Publisher: Multidisciplinary Digital Publishing Institute
9. Malaperdas, G.D.: Returning to the past for the revival of the future: analyzing the value of old maps for the preservation of cultural heritage. In: *Digital Approaches to Cartographic Heritage Conference Proceedings* (Jan 2022). <https://www.academia.edu/96676953>
10. Midura, R.: Itinerating Europe: early modern spatial networks in printed itineraries, 1545–1700. *J. Soc. Hist.* **54**(4), 1023–1063 (2021)
11. Petrulevich, A., Skovgaard Boeck, S. (eds.): *Digital spatial infrastructures and worldviews in pre-modern societies*. Arc Humanities Press (2023). <https://doi.org/10.17302/CDH-9781802700794>, <https://library.oapen.org/handle/20.500.12657/62540>, accepted: 2023-04-24T14:00:30Z
12. Reckziegel, M., Wrisley, D.J., Hixson, T.W., Jänicke, S.: Visual exploration of historical maps. *Digital Scholarship Humanities* **36**(Supplement.2), ii251–ii272 (Oct 2021). <https://doi.org/10.1093/llc/fqaa059>, <https://doi.org/10.1093/llc/fqaa059>
13. Scheidel, W.: *ORBIS: The Stanford Geospatial Network Model of the Roman World* (2015). <https://doi.org/10.2139/ssrn.2609654>

14. Spadi, A.: Digitales forum romanum: un progetto per la ricostruzione storico-scientifica in 3D del Foro Romano. *Umanistica Digitale* (7) (Dec 2019). <https://doi.org/10.6092/issn.2532-8816/9330>, <https://umanisticadigitale.unibo.it/article/view/9330>, number: 7
15. Terpstra, N., Rose, C.: *Mapping Space, Sense, and Movement in Florence: Historical GIS and the Early Modern City*. Routledge (2016)